



CE 566: Disaster-Resilient Mountain Road Engineering

Course Code: CE 566

Course Name: Disaster-Resilient Mountain Road Engineering

L-T-P-C: 2-1-0-3

Prerequisites: N/a

Intended for: UG & PG students

Mutual Exclusion: N/a

Distribution: Discipline elective

Approval: 60th BoA

Course Preamble

Understanding natural hazards, the risks they pose to road infrastructure and its users, and the design of resilient infrastructure is crucial for all stakeholders working in road infrastructure in hilly regions—including engineers, policymakers, and planners. Efficient and effective communication and coordination between all stakeholder groups is essential to ensuring adequate resilience. This course will bridge the gap among the various hill road stakeholders by improving awareness of different roles and responsibilities and introducing common frameworks and terminologies. The course offers foundational knowledge on risk assessment and risk reduction approaches, management and maintenance strategies, and current policy for design, construction, and resilience of hill roads. Real-world case studies from diverse stakeholder perspectives are integrated to ensure participants gain practical, experience-based insights.

By the end of the course, participants will have gained understanding of the key terminologies, frameworks, and concepts in hazards, risk, and resilience for hill roads. This knowledge will enable participants to have effective communication and coordination with other stakeholders. They will also gain practical skills to implement this knowledge effectively in real-world scenarios.

Course Modules

- **1. Introduction to hazards, risk, and resilience for hill roads:** Understanding mountain dynamics (tectonics, rainfall, geology, geomorphology, and slope aspect); Road classification in hilly regions; Hazard types and definitions; Hazard-infrastructure interaction; Introduction to multi-hazards; Impact of climate change on hazards in hilly terrain; Introduction to hazard identification and assessment; Introduction to risk and resilience terminology; Introduction to engineering solutions. (14 hours)
- **2. Concept of risk assessment:** Risk terminology; Hazards Elements at risk/Exposure; Vulnerability; Overview of risk assessment types; total risk, risk tolerance; Strategies to characterize risk based on road characteristics; Risk communication. (6 hours)
- **3. Risk reduction:** Geological, geomorphological, and engineering assessment, Mitigation (including drainage implementation); Role of technology such as early warning and monitoring systems; Indigenous and nature-based solutions; Community participation, education, and preparedness; Accounting for climate change. (6 hours)
- **4. Hill road operation and maintenance:** Role of stakeholders; Operation, monitoring and preventative maintenance; Reconstruction/ rehabilitation; Drainage management; Early warning systems; System recovery. (6 hours)

- **5. Network resilience and cost-benefit analysis for risk reduction:** Prioritization of action on assets; Planning strategy to maintain the road network; Identification of indicators to represent costs and benefits. (6 hours)
- **6. Policy and governance:** Existing policy and guidelines for design, construction, maintenance, and resilience for different categories of roads; Funding and financing mechanisms; Climate goals for resilient roads. (4 hours)

Main Text Books

1. Blaikie, P., Cannon, T., Davis, I., & Wisner, B. (2014). *At risk: natural hazards, people's vulnerability and disasters*. Routledge, London.
2. Fuchs, S., & Thaler, T. (2018). *Vulnerability and resilience to natural hazards*. Cambridge University Press, Cambridge.
3. Hearn, G. (2011). *Slope Engineering for Mountain Roads*. Geological Society Engineering Geology Special Publication No. 24. The Geological Society, UK.
4. Infrastructure Resilience Division of ASCE. (2021). *Hazard-resilient infrastructure: Analysis and design*. MOP-144, Edited by B. M. Ayyub, Reston, VA: American Society of Civil Engineers.
5. Kramer, S. L. (1996). *Geotechnical earthquake engineering*. Prentice Hall, New Jersey.
6. Semien, J., & Nance, E. (2022). *Hazard mitigation training for vulnerable communities: a KAPS (knowledge, attitude, preparedness, skills) approach*. Routledge, New York.

Additional Reading

Journal papers and reports in the area of disaster risk reduction, hill road construction, and natural hazards